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A Probabilistic-Driven Approach for Early Design Quality Risk and Crux Identification Using Non-Markovian Stochastic Petri Nets

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Abstract - Stochastic Petri Net (SPN) is a powerful and widely used approach for modelling, deadlock detection and safety analysis of stochastic processes in complex stochastic systems. However, applying this method is rarely seen in formal Risk Analysis, especially for quality risks, which refer to the system's capability and performance to fulfil its objectives on time. This paper investigates the potential of applying Risk Analysis in the early conceptual design stages by modelling the design problem using non-Markovian SPN as a formal analytical tool. The use of SPN offers several advantages, such as high flexibility and strong compatibility with statistical methods such as performance analysis with semi-Markov models, as well as sensitivity analysis and uncertainty analysis. The proposed method enables designers to address quality risks quantitatively and focus explicitly on the design crux problem - the key quality issue directly affecting the design's success. Preferred high-level solutions obtained throughout the Risk Analysis are evaluated using Monte Carlo simulations as initial decision-making insights. A case study of the concept development of a remote maintenance system for the In-Bioshield area of the DEMO fusion power plant is presented to demonstrate the method's applicability. Initial results showed potential in identifying quality risks, addressing key factors contributing to the design problem, and finding optimal design specifications in the early stages.

Keywords: *conceptual design, quality risks, stochastic Petri Net, non-markovian stochastic Petri Net, sensitivity analysis, uncertainty analysis, semi-markov processes, performance analysis, risk analysis.*

1. Introduction

Engineering design, an integral part of product development, focuses on solving design problems through an iterative, systematic and creative approach from a technical perspective. Typical engineering design methodologies (Pahl et al., 2007; Ulrich & Eppinger, 2012) follow structured processes that begin with analysing design requirements, clarifying tasks, and progressing through conceptual, embodiment and detailed design stages. Among these processes, the conceptual design stage is considered the fundamental and core of the product, defining its functionality and overall value (Andreasen et al., 2015; Komoto & Tomiyama, 2012; Zheng et al., 2016).

In the conceptual design stage, the design context and requirements are thoroughly analysed to identify the crux of the problem and the system's overall behaviour. A design crux problem refers to the central focus of a design task, emphasising key customer requirements without distraction from non-essential details (Pahl et al., 2007). It is identified by abstracting and synthesising the design requirements as an overall function. The overall function is decomposed into subsystems and formed into a function structure diagram representing their functional interrelationships. Solutions and working principles are searched for each subsystem and combined into Concept Variants (CVs). The principle solution is selected through the evaluation and benchmarking of CVs. Fig. 1 summarises a typical conceptual design and four key characteristics: problem-solving, searching for solutions, concept development and decision-making.

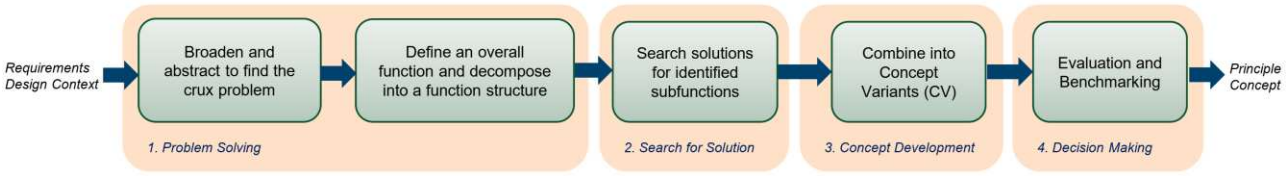


Fig. 1 Summary of the conceptual design process and its four key characteristics.

It is evident that the essence of the conceptual design is a sequential and iterative process in which the early stages significantly influence design directions, shape and quality of the concept in later design steps (Jonassen, 2000). This observation has been widely agreed upon in recent studies, where most discussions focus on the importance, comprehension and implementation of problem definition and solving based on the inputs of the design problem (Jonas, 1993; Smith & Browne, 1993). Andreasen (Andreasen et al., 2015) indicated that a poor understanding of the design context and problem-solving in the initial stages can lead to major risks, potentially failing a product to fulfil the

customer's expectations and market needs. In another aspect, Pahl and Bietz (Pahl et al., 2007) argue that the rush in problem-solving stages should be avoided as it implies the risk of overlooking better and more economical solutions, leading to longer design time and cost. This tendency risk is slightly related to confirmation bias - a propensity to search and interpret information to support one's beliefs, which has been investigated and indicated to impact concept generation and evaluation significantly (Arnott, 2006; Hallihan et al., 2012). Both authors suggest that "broadening the problem formulation" should be systematically divided into a sequence of processes and illustrate this application through Krick's notable example (Dodge, 1969).These discussions show that the broadening problem formulation at the problem-solving stage is important in conceptual design and should be perceived appropriately. Although this step can be improved by decomposing design problems into systematic sequential processes to facilitate abstractions, it relies heavily on intuitive and qualitative assessment, which can be vulnerable to early design risks. In this article, "quality risk" refers to early design risks related to the performance metrics the design system must achieve at optimal efficiency and support the finding of the design crux problem. These metrics impact searching for solutions and decision-making characteristics for conceptual design. Therefore, broadening problem formulation and addressing quality risks requires a quantitative approach at the early stages of the conceptual design to enhance problem-solving, identify the crux problem, and key performance measures. Thus, due to the iterative nature of engineering design, the method should not be constrained to the initial stages only. Instead, it should be systematically integrated throughout the conceptual design, including the final evaluation and decision-making phases, as a quantitative tool for assessing and benchmarking concept performances.

The strengths and requirements of using quantitative methods for Risk Analysis (RA) instead of qualitative approaches have been confirmed and emphasised in recent decades in the RA community (Anthony (Tony)Cox, 2008; D. Hubbard & Evans, 2010; D. W. Hubbard, 2020). This shift is driven by high accuracy and is less influenced by users' confirmation biases or current moods than traditional risk labelling and scoring frameworks. However, in the early design stage, quantitative RA methods face several challenges as the product form has not yet been fully defined. This is obvious because most information is only design context and requirements at this stage. One feasible solution to overcome this challenge is to model the design problem as a sequence of processes, analyse it from a mathematical perspective, and solve it to find optimal specifications as part of the design goal. This idea can be achieved by formalising system processes as discrete events and performance uncertainties as random variables. Naturally, this approach leads to stochastic processes, where techniques such as Stochastic Petri Nets (SPN) with statistical methods such as Performance Analysis based on semi-Markov (SM) models, Sensitivity Analysis (SA) and Uncertainty Analysis (UA) are applicable.

SPN is built based on Petri Nets (PN). Traditional PN, a popular mathematical modelling language developed by Carl Adam Petri (P. Carl Adam, 1962), offers a graphical representation built on a mathematical foundation, making it easier to visualise sequential or concurrent processes and simplifying the complexity of mathematical solving in early design. The use of PN has been gaining popularity in recent decades for various applications, such as modelling logistic processes (Behzad et al., 2020) or maintenance processes (Benzoni, Introini, et al., 2025; Ponciroli et al., 2016; Sheng & Prescott, 2019). One notable study is the potential application of PN in the conceptual design of manufacturing systems (Silva & Teruel, 1997), in which the PN's were described to exhibit high compatibility with the conceptual design stage through interrelationships between subsystems and their qualitative impact on the overall system behaviours. These characteristics allow users advantages in equipment selection, allocating resources and system planning. Indeed, the use of PN in modelling design problems with a strong process-oriented tendency has been proven by Benzoni, et al., 2025, which shows that PN can serve as a tool to compare the performance of different concepts and advantages in identifying key processes.

However, PN has certain limitations, as the firing time of transitions is typically fixed at a final value, restricting its application to problems where transition times are known or predefined as fixed values. This characteristic, however, is not practical in some circumstances, such as in conceptual design, where system parameters are yet fixed at a constant value; instead, they are estimated under a possible range value in design consideration. In contrast, the SPN model fires delays of transitions as random variables, allowing the representation of variance in process time and uncertainties using statistical methods (Sibois, 2016). This property supports integrating sensitivity and performance analysis using semi-Markov models. SM is the generalisation of Markov processes by allowing arbitrary transition-time distributions, which facilitates the inclusion and representation of uncertain or risky processes in the sequential process. Such approaches enable more realistic system behaviour modelling in early design, especially when uncertainties or risks exist. The outcomes of these studies can be comprehensively re-evaluated using Monte Carlo simulations, which are also supported by many SPN modelling applications.

From these motivations, this article presents a method of integrating Risk Analysis at the initial phase of conceptual design within Pahl's framework (Pahl et al., 2007) by expanding the first step of the method and applying SPN as a modelling tool to structure the design problem, establishing an integration foundation with Sensitivity Analysis and Uncertainty Analysis to identify potential system quality risks and facilitate the finding of the crux of the design. The paper is structured into 4 sections: Section 2 first provides a formal definition of quality risk and a brief review of the theoretical foundations, including SPN, performance analysis using semi-Markov models, Sensitivity Analysis and Uncertainty Analysis. In Section 3, these theoretical backgrounds are analysed for their applicability and relevance to

engineering design, forming the basis for the synthesis of a new proposed method introduced in this section. The practical use of the method is demonstrated in Section 4 through the development of Remote Maintenance Equipment (RME) for the In-BioShield of the DEMO fusion power plant. Discussions and conclusions are drawn in Section 5.

2. Review of Theoretical Backgrounds

2.1. Quality Risk in Engineering Design

Risks refer to events or outcomes that have undesirable impacts. RA aims to identify what causes risks and how to prevent or mitigate possible consequences. Originating in the 18th century from finance and insurance, RA initially focused on profits or momentary loss (D. W. Hubbard, 2020), but has since expanded to the concern of schedule-related risks in business models based on analysing event logs (process mining) (Camargo et al., 2020; Rogge-Solti & Weske, 2015), known as performance analysis.

In product and system design, risks have a broader term originating from technical aspects, such as system reliability and high-level uncertainties, including development time, market conditions, and profitability (de Weck et al., 2007). In process-oriented systems (e.g., manufacturing or logistics), these risks often concern about processes and system performance. These priorities strongly relate to performance analysis, similar to RA in business models. A poor performance often arises from multiple factors, such as machine breakdown, scheduling, and quality failures (L. A. Shah et al., 2017). Therefore, it is essential to define system performance risks and evaluate them in the early product life cycle (L. Shah et al., 2012). A recent study has identified risks that may emerge during early-stage system design, some closely linked to performance issues (Truong & Brace, 2024). As shown in Fig. 2, quality risk in engineering design is defined as risks from poorly designed strategies or system configurations that hinder optimal performance in achieving the overall task and arise as consequences of failure modes or situational risk factors.

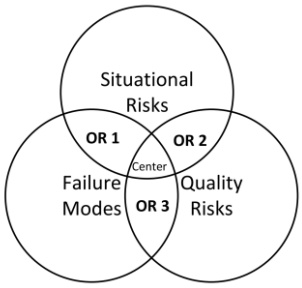


Fig. 2 Technical Risk Model in System Design with 3 Overlapped Regions (OR), adapted from (Truong & Brace, 2024).

The term Quality Risks (QR) in this article refers to performance risks that might impact the system's efficiency and ability to complete the overall process on time, in the engineering design of highly process-oriented systems. The analysis of QR in the early conceptual design focuses on formulating design problems as a general process and identifying the riskiest factors that could impact the system's performance and its relations to uncertainty scenarios related to failure modes or situational risks. These factors can be sub-processes or key design specifications and are observed as a part of design crux problems through the process view. The risk assessment of QR compares quantitatively possible high-level solutions that come up at this stage based on RA results, providing insights on the product's high-level specifications as additional requirements to fulfil the requirements of the products.

2.2. Stochastic Petri Net and Non-Markovian Stochastic Processes

The PN approach defines high-level and generalises the system processes to allow different models to be tested and easily expanded. PN is a mathematical modelling language that offers a graphical notation for stepwise, concurrent, and parallel processes that suit the requirements of the QR analysis, in particular, thanks to the discrete nature of the analysis. In the PN, the system modelled can be assimilated to a "token" that tracks the task it is currently doing inside a general process. Indeed, the developed PN model uses discrete places and transitions to simulate the process analysed, where places are akin to the "system state" concept. Transitions represent events that lead tokens (the system) from one state to the next. The PN is modelled in Modelica in this work, leveraging the PNlib library (S. Proß et al., 2024). Modelica, an object-oriented simulation language (Fritzson, 2014), increases the adaptability of the developed models, enabling possible reuse instead of development from scratch. For example, a simple Input/Output (I/O) system is represented in Fig. 3 where the system in the input state (placed with I) is taken to the output state O from the transition T. In this case, the system is represented by the PN token that starts from the Input state (place I) and goes into the output state (place O). The transitions are usually represented by a firing time, which accounts for the duration of the process. The different elements of the PN is then connected via "arrows" called arcs.

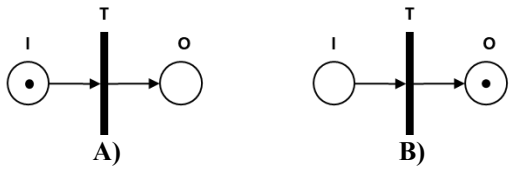


Fig. 3 Petri Net representation of a simple I/O system (adapted from (Benzoni, et al., 2025)).

In PN, the firing process of the transition can also be instantaneous, where no duration or delay is associated. This latter transition can be used to model the scarcity of resources or to generate tokens to enable a parallel branch, which could represent a system that starts operating. Indeed, PN's main scope is to simulate concurrent parallel systems with multiple possible outcomes (or inputs). The instance where a place in the PN is connected to multiple input or output transitions is also called "conflict".

Fig. 4 shows an example of a conflict (an output one) for a place with two output transitions. The conflict arises since the token can be consumed only by one of the two transitions (the system can only occupy one state). Thus, to establish whether either of the two branches shown in the figure (represented by T1 or T2) receives the token, a priority rule must be introduced. Priority to the transition can be attributed in several ways, such as pure order (where T1 will always precede T2) or by including a probability to the branches. The introduction of rules is required to establish the priority of the input transitions. The use of this conflict effectively captures worst-case scenarios in RA. Specifically, PN with conflict representation can be employed to model the design problem as a step-by-step operational process for the design system to fulfil its requirements.

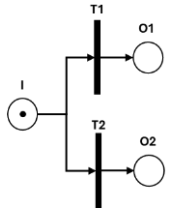


Fig. 4 Output conflict representation (adapted from (Benzoni, et al., 2025)).

However, modelling the system using classical PN, where transitions are either immediate or have fixed firing times, is insufficient to capture variability in transition behaviour. Instead, Stochastic Petri Net (SPN) address this limitation by modelling firing times as random variables. According to the definition of SPN (Falko Bause & Pieter S. Kritzinger, 2002; Michael Karl Molloy, 1981; S. Natkin, 1980), an SPN with m transitions can be seen as a standard PN augmented with an additional set $\Lambda = (\lambda_1, \lambda_2, \dots, \lambda_m)$, where each λ_i denotes the transition rate of the transition t_i. This means that the firing times of each transition t_i follow an exponential distribution defined by the cumulative distribution function (CDF):

$$F(x) = 1 - e^{-\lambda t} \tag{1}$$

By modelling firing times as exponential random variables, an SPN exhibits the properties of a continuous-time Markov process with a discrete state space. A process is considered Markovian if its future state depends only on the current state (Markov property), and in continuous time, memorylessness (memoryless property) additionally implies that the sojourn time in each state follows an exponential distribution. When all transition time in an SPN follows exponential distributions (satisfies memoryless property), the SPN can be classified as a Markovian SPN.

Modelling real-world processes using a strictly Markovian SPN can be observed as unrealistic, as it constrains the execution time (firing time) of each subprocess to the exponential distribution. In statistical modelling, exponential distributions are typically used to describe the time intervals between consecutive events (e.g., the time interval between machine failures). However, the use of this distribution type is still remains common in machine scheduling models due to mathematical convenience rather than empirical accuracy (Trietsch et al., 2012). To better reflect real-world variability, many researchers have proposed using alternative long-tail distributions to capture the variability and delay in process durations more accurately. For example, Baker and Trietsch (Baker & Trietsch, 2009) recommended using lognormal or Parkinson distributions in modelling stochastic activity times. Other empirical studies also support this approach, as lognormal distributions can be used to describe surgical durations (May et al., 2000; Strum et al., 2000) and drilling operations (Premachandra & Gonzalez, 1996). Such findings highlight that real-world processes often follow diverse, non-exponential distributions. Moreover, adopting long-tail distributions enhances risk analysis by allowing a better assessment of rare but critical events represented in the tail of the distribution's process.

These limitations led to several extensions and developments of Non-Markovian SPNs, such as Generalised Stochastic Petri Nets (GSPN) (Ajmone Marsan et al., 1984) and Stochastic Reward Nets (SRN) (Ciardo et al., 1993), which relax the exponential assumption and support non-Markovian behaviours. Therefore, this work adopts a non-Markovian SPN

as a generalised framework for modelling a detailed step-by-step sequence for a design system to complete its required tasks. The SPN model incorporates conflict branches to represent worst-case scenarios with branching probabilities. The firing time of each transition follows an arbitrary probability distribution that reflects the statistical characteristics of the corresponding process.

2.3. Performance Analysis using Semi-Markov Models

Performance analysis is one of the key objectives of process mining, which uses key indicators to examine a system or process's performance (van der Aalst, 2016). One key performance metric is the overall process time (known as throughput, lead or flow time), which measures the duration from the beginning to the end of a process. Examining the overall process time allows analysts to identify process bottlenecks, assess operational performance, and gather evidence to diagnose and optimise issues.

The concept of "overall process time" used in performance analysis closely aligns with the definition of QR introduced in this article. The main idea is to convert the non-Markovian SPN into a corresponding irreducible Markov chain, then apply performance analysis with SM models to evaluate worst-case scenarios. An irreducible Markov chain means that every state is reachable from at least one other state in the network. While the original non-Markovian SPN captures concurrent processes, resource constraints, and conflicts, the Markov chain summarises and maps these elements as a high-level risk network, where states represent worst-case scenarios. By leveraging SM models, the network's throughput time can be estimated analytically based on this Markov chain. This approach is feasible because non-Markovian SPN and semi-Markov models share similar characteristics (Kalenkova et al., 2025).

This study's theoretical framework for performance analysis using SM models is primarily based on the express calculation outlined in the article (Kalenkova et al., 2025). The calculation requires the SPN to be transformed into an irreducible Markov chain by connecting the initial state s to the end state e. This connection makes the SPN a closed-loop network, established through an immediate transition (with mean time = 0) and a transition probability of 1 between the initial and final states. The SM model has the same continuous-time property (Sheldon Ross, 2014), similar to the SPN, and is characterised by a stochastic matrix P. The stochastic matrix P contains non-negative real numbers, each representing a transition probability $p_{i,j}$ from state i to state j in the Markov chain ($1 \leq i,j \leq m$). By solving $\pi P = \pi$, where $\pi = (\pi_1, \pi_2, \dots \pi_m)$ and $\sum_{k=1}^{k=m} \pi_k$, one can derive the steady-state (limiting) probabilities π_i for each state, representing the long-run probability of the system being in state i as $t \to \infty$ (Gagniuc, 2017). The process can be optimised by minimising the highest limiting probabilities or allocating additional resources to reduce the overall process meantime. The overall meantime from the beginning state s to the end state e of an SM model can be calculated as follows (Kalenkova et al., 2025):

$$\mu = \frac{1}{\pi_s} \sum_{\substack{i=1, \\ i \neq e}}^{m} \pi_i \mu_i \tag{2}$$

Where π_i is the steady-state (limiting) probability of the state i ($1 \leq i \leq m$ and $i \neq e$) and μ_i is the meantime of transitions from the state i. Equation (2) is an express calculation based on the meantime, not considering the shape and type of transition distributions.

Fig. 5 illustrates an example of this idea by mapping a non-Markovian SPN to the irreducible Markov chain. The initial SPN network has an output conflict representing a worst-case scenario in two branches – the first branch (state corresponding to the place S2) with T1.1 (mean transition time $\mu = 3$) and T1.2 transitions ($\mu = 1$) with a probability of 0.6 and branch two (state S3) with T2.1 ($\mu = 1.5$) and T2.2 ($\mu = 1$). The unit of transition time in this example is hours. Both cases lead to the end state S4 of the process. The original and open SPN is formed into a closed network by introducing an immediate transition T3 with mean time $\mu = 0$. The SM model of the example in Fig. 5 has the following stochastic matrix:

$$P = \begin{bmatrix} 0 & 0.6 & 0.4 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \tag{3}$$

The limiting probability can be derived as $\pi_1 = \pi_4 = \frac{1}{3}, \pi_2 = 0.2, \pi_3 = \frac{2}{15}$. By applying equation (2), the expected process overall time can be determined as (note that π_4 is excluded since it is the end state – e):

$$\mu = \frac{1}{\pi_s} \sum_{\substack{i=1, \\ i \neq e}}^{m} \pi_i \mu_i = \frac{1}{\pi_1} [\pi_1 \cdot (0.6 \cdot 3 + 0.4 \cdot 1.5) + \pi_2 \cdot 1 + \pi_3 \cdot 1] = 3.4(hours) \tag{4}$$

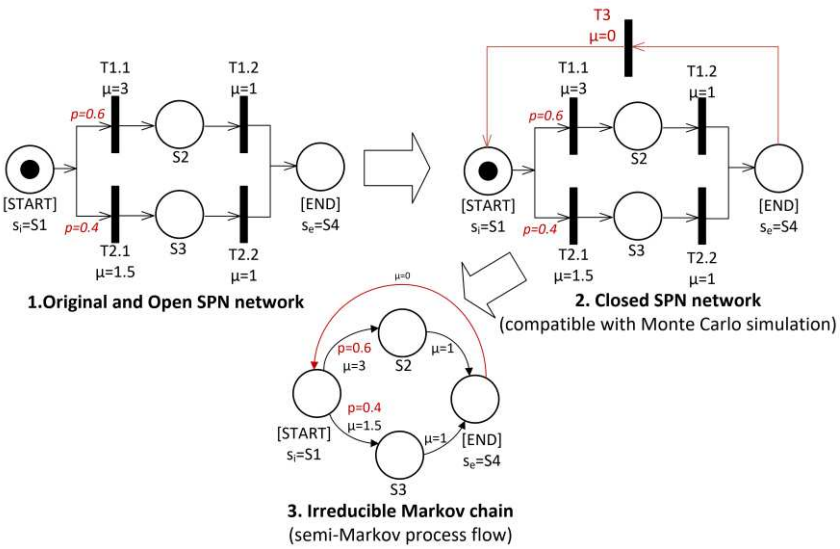


Fig. 5 An example of an open non-Markovian SPN network translated into an irreducible Markov chain.

In this example, the above equation (4) is computed using fixed numerical values. Suppose we generalise the uncertainty parameters, such as the probabilities of conflict branches, into symbolic variables. In that case, the overall mean time from equation (2) becomes an interrelationship function between uncertainty variables (input) and the system's performance (output – known as QR).

2.4. Sensitivity Analysis and Uncertainty Analysis

Consider a conceptual design as a mathematical model, where the overall mean time (Y) required for the system to complete a given task serves as a QR index of interest, measured in units of time. Then, the design can be treated as a black box function that transforms p number of input variables $X = (x_1, \dots, x_p)$ into the output Y, as follows:

$$Y = f(X) = f(x_1, \dots, x_p) \tag{5}$$

The input variables $X = (x_1, \dots, x_p)$ here represent the target specifications of the system being designed. These specifications may include capabilities, such as the speed of a subsystem or the possible distance the system is expected to perform, depending on the design context. Equation (5) implies that design concepts map each input specification set to a different estimated performance. Finding design crux problems in the context of QR means determining the most important input variables (i.e. key design specifications) affecting system performance Y. Analysing and minimising QR involves expanding the model to include additional uncertainty variables (where applicable) and optimising the possible values of the key input variables to ensure acceptable system performance aligned with the design concept. Addressing QR at this early stage means working with high-level product specifications rather than detailed functional architectures. This direction aligns closely with Pahl's black box modelling approach(Pahl et al., 2007), where the system is viewed as an overall function that maps inputs to outputs.

When design complexity increases, the number of inputs p is larger; the relationship between input and output variables in equation (5) is difficult to interpret or predict. Thus, input variables in practice are often subjected to noise and stochastic events, which are known as uncertainties, affecting the model's output variances. Therefore, understanding the influence of input variables on the output is essential for effective QR analysis. SA and UA are the two widely used approaches to analyse such models effectively.

SA is the study of how input factors influence the output variables of a mathematical model. In other words, it examines the effect of X on Y in equation (2) to identify the most impact factors. There are numerous methods for performing SA, such as visual analysis, one-at-a-time (OAT) or Morris. However, within the scope of this study, SA is used as a supporting tool to facilitate finding design crux problems as a part of RA during the conceptual design phase. Therefore, this paper does not delve into the full theoretical background of SA. Instead, we briefly overview a specific and representative approach: the variance-based SA based on the computation of Sobol indices, which is applied in the case study presented in Section 3.

The Sobol indices belong to the variance-based methods, a global SA method for solving nonlinear models. In Sobol's (Sobol, 1993; Sobol′, 2001) approach, the output function Y in Equation (5) is decomposed into summands of increasing interaction order.

$$Y = f_0 + \sum_{i=1}^{p} f_i(x_i) + \sum_{i<j}^{p} f_{ij}(x_i, x_j) + \cdots + f_{12\ldots p}(x_1, \ldots, x_p) \quad (6)$$

Where f_0 is an expected value of the output; the term $\sum_{i=1}^{p} f_i(x_i)$ represents the variance influenced by individual single input variables $x_1, \ldots, x_p$; the term $\sum_{i<j}^{p} f_{ij}(x_i)$ is the variance caused by interactions between 2 variables x_i and x_j (second-order interaction); $f_{12\ldots n}(x_1, \ldots, x_p)$ is the variance due to the higher-order interactions among p variables. This number of interactions considered between input variables is denoted by s. Each component in equation (6) satisfies the following orthogonality condition:

$$\int_0^1 f_{i1\ldots is}\,(x_{i1}, \ldots, x_{is}) dx_k = 0 \text{ with } k = i_1, \ldots, i_s \quad (7)$$

To quantify the contribution of each input or combination of inputs to the total output variance of Y, Sobol's indices are computed. The first-order Sobol index S_i measures the individual contribution of the variable x_i, while the total-effect index S_T describes both its individual effects and all interaction effects involving x_i. For example, for a model with three inputs x_1, x_2, x_3, the total-effect index of x_1 includes its direct and interactive contributions as follows:

$$S_T = S_{T1} + S_{T12} + S_{T13} + S_{T123} \quad (8)$$

The computation and application of Sobol's indices offer an effective approach to handling nonlinear problems, as the contribution of each input variable to the output variance is quantified. The calculation of these indices is widely supported by tools such as the SALib open-source library (Herman & Usher, 2017; Iwanaga et al., 2022) in Python.

In addition to SA, UA typically aims to characterise the distribution of the output function Y using various statistical representations such as summary statistics, probability density functions (PDFs), or cumulative density functions CDF. Its main goal is to quantify the likelihood of uncertain values in the output Y. In the context of QR analysis in the conceptual design, UA can be utilised to assess risk by comparing different design configurations. However, in cases where input variables exhibit only minor changes, the differences in output Y between design configurations tends to be minimal. Deciding which configuration is better becomes more challenging, especially since the output is shown as a distribution, not just a single value. Non-parametric effect size measures like Cliff's Delta can facilitate this decision-making.

Cliff's Delta (δ) (Cliff, 1993, 2014) is used to quantify the degree of difference between any two arbitrary distributions (x_{i1} and x_{i2}), especially when they are not normally distributed. It measures the extent of the difference by calculating the non-overlapping area between the two as follows (Kane & Esther, 2024) (n_1 and n_2 are the sample size of x_{i1} and x_{i2}):

$$\delta = \frac{\#(x_{i1} > x_{i2}) - \#(x_{i1} < x_{i2})}{n_1 n_2} \quad (9)$$

Calculation results from equation (9) give the value of the Cliff's Delta, which ranges from -1 to 1, where:

- $|\delta| < 0.15$: No significant difference between the distributions
- $0.15 \leq |\delta| \leq 0.33$: Small differences and effects between the distributions
- $0.33 \leq |\delta| \leq 0.47$: Medium differences and effects between the distributions
- $0.47 < |\delta|$: Large differences and effects between the distributions

3. Methodology

3.1. Leveraging SPN, Performance Analysis, SA and UA as an RA Methodology in Early Conceptual Design

The proposed framework extends Pahl's design method (Pahl et al., 2007) by applying RA at the earliest stage of conceptual design, particularly between the "Requirement list" and the "Abstract to identify the essential problems", as shown in Fig. 6. In this proposed method, the design problem is represented as a sequence of discrete events and modelled using non-Markovian SPN and subsequently transformed into irreducible Markov chain. The design problem is further formulated as a mathematical problem using Equation (5), comprising design and uncertainty parameters as X input. SA is employed to investigate the relationships and impacts of input design parameters on system performance. The irreducible Markov chain is utilised for performance analysis using SM models to examine how worst-case scenarios affect the system through uncertainty parameters. Findings from these two analyses are synthesised to perform Monte Carlo simulations and UA to determine the optimal design specification. The overall purpose of the approach is to identify QR as a comprehensive understanding of how design specifications influence system performance and consider the

likelihood of undesired scenarios. The outcome of the method is an improved understanding of the QR on system performance and preliminary preferred design specifications to mitigate these risks.

In particular, the framework consists of two main phases, as shown in Fig. 6: broadening the problem formulation and abstracting it to identify the essential problem. In the first phase, the design context and requirements are examined and formulated as a mathematical problem, incorporating discrete events describing how the system operates to meet the design requirements. An SPN network is then modelled to conduct two types of analysis:

- **Sensitivity-based analysis:** identifies the most critical high-level process in the system and the design specifications that impact system performance using SA.
- **Performance-based analysis:** constructs based on the results of the sensitivity-based analysis and includes potential risks affecting the system as worst-case scenarios. This analysis is based on performance analysis with semi-Markov models to refine and broaden the design problem.

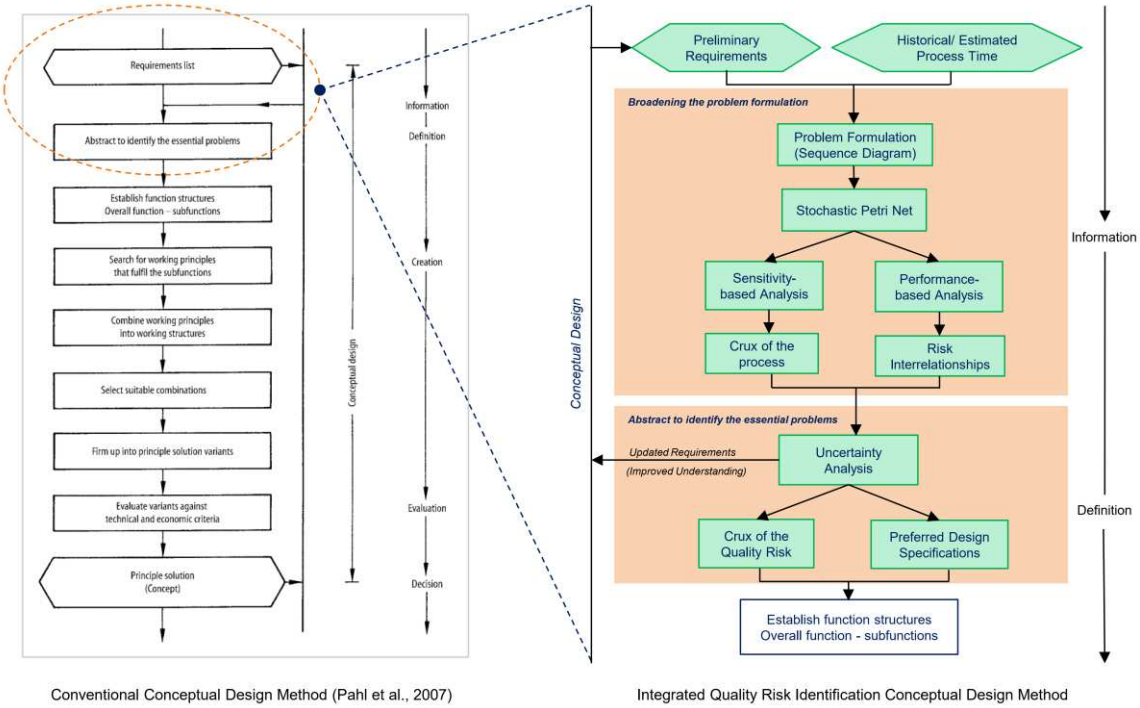


Fig. 6 Improved Conceptual Design Methodology with early QR identification (right) built upon the original conceptual design methodology adapted from (Pahl et al., 2007) (left).

From the results of the two analyses, the method moves on to the second phase to identify the preferred design specifications that balance QR and design trade-offs. The following sections describe these steps in detail (the detailed method for each phase is depicted in Fig. 7).

3.1.1. Problem Formulation and Sequence Diagram

A system sequence diagram is systematically constructed based on preliminary requirements and design context. This diagram provides a step-by-step breakdown of the small processes the system needs to operate to complete the overall task. There are no strict guidelines for structuring this diagram, and designers are flexible in generating creative processes as long as the system fulfils the requirements. Analysing the design problem regarding processes provides an initial understanding of the system's main functions and behaviours and later supports establishing the function structure.

3.1.2 Sequence Diagram to Stochastic Petri Net

Once the sequence diagram has been constructed, the design problem is then translated into a non-Markovian SPN. Each process is represented through a stochastic transition. At this stage, the design task is conceptualised as a black box function (equation 5), where inputs are:

- **Design parameters:** represent high-level product specifications that are considered. These parameters contribute directly to system performance or the time to complete each process on the sequence diagram.
- **Uncertainty parameters:** Uncertainty factors arise from the problem's context, such as worst-case scenarios from situational risks or failure modes that the system might encounter. These factors can be the branching probabilities of output conflicts in the SPN network or transition times.

The output of the black box is the system's performance (expected process overall mean time). The objective of the QR Analysis is to:

- Identify key design parameters (crux parameters) that impact the system performance most.
- Understand how uncertainty parameters influence the system performance
- Defining acceptable design parameters to ensure system resilience under these scenarios

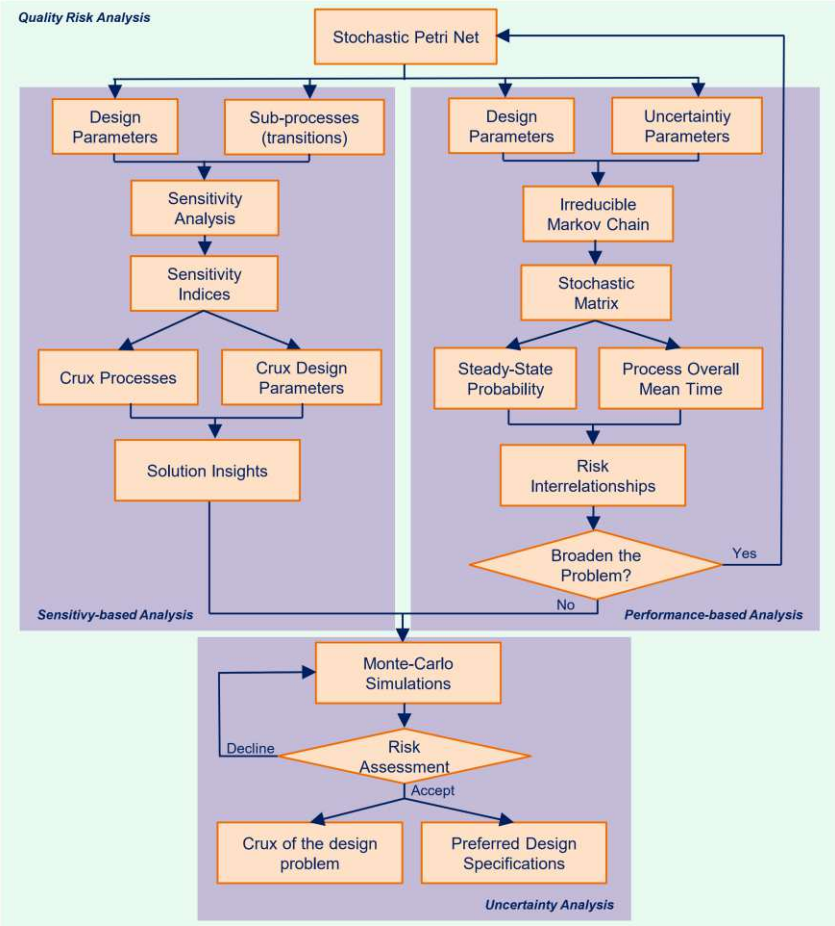


Fig. 7 The detailed method for each highlighted phase in Fig. 6, as a synthesis of sensitivity-based, performance-based and uncertainty analysis.

3.2. Sensitivity-based Analysis

Sensitivity-based analysis evaluates the impact of input design parameters on system performance using SA by modelling the design as a black box function following Equation (5) with inputs as design parameters and the output as the system's performance. Different SA techniques can be used depending on the design complexity and desired precision levels. From the SA results, the most critical design parameters and transitions impacting the SPN's completion time can be identified, equivalent to determining the crux design parameters. Recognising the crux design parameters and key processes prevents the design strategy from drifting into inefficient solutions.

3.3. Performance-based Analysis

While sensitivity-based analysis focuses on design parameters, performance-based analysis examines how uncertainty parameters impact the performance of the design problem. At this stage, the SPN is translated into a closed network with an irreducible Markov chain, where risk scenarios are captured through probabilistic transitions between states. Worst-case scenarios are represented as parallel branches with associated probabilities of occurrence within the SPN. The SM model of the Markov chain is then solved and analysed to determine:

- **Steady-state probabilities:** indicate the system's likelihood of entering these worst-case scenarios (states). These are derived using basic SM equations.
- **Interrelationships between process overall mean time and uncertainty parameters**: providing insights into how uncertainties influence the design problem under the interrelationship function based on equation (2). Note that uncertainties are formulated as symbolic variables, and equation (2) enables quick estimation based on expectations (variance in the stochastic distribution is not considered).

Once the interrelationships between risks and system performance are understood, designers can define acceptable uncertainty values as reliability requirements for the design, depending on the design context. This analysis can be performed sequentially and iteratively to broaden the design problem.

3.4. Uncertainty Analysis

The QR of the system is assessed through the uncertainty analysis and can be evaluated using Monte Carlo simulations on an SPN network based on acceptable values derived from the sensitivity-based and performance-based analyses, utilising the closed SPN network:

- **Sensitivity-based analysis:** reasonable threshold values for design parameters to optimise the system performance
- **Performance-based analysis:** appropriate uncertainty values to be considered in simulations

It is noted that there is always a trade-off involved in selecting these values; particularly, better design input parameters do not always guarantee better performance or ease of implementation in further product development. Simulation results are then interpreted via UA using statistical methods to compare design parameter sets and support decision-making. Once a consensus is reached within reasonable design parameter values and acceptable uncertainty, the crux of the design problem can be summarised, and these parameters can be used as objectives for subsequent design stages.

4. Case Study: Development of RME for Fusion Power Plant

The proposed method's practical applicability is researched through its implementation in an ongoing project aimed at developing a conceptual design for an automated remote maintenance (RM) system for the In-Bioshield (In-BS) area of the EU-DEMO fusion power plant. This particular case study topic was also implemented in previous work (Benzoni, et al., 2025) but with a stronger focus on the transportation and deployment aspects to perform non-destructive testing inspection.

4.1. Problem Formulation and Preliminary Sequence Diagram

In the context of this design problem, the system to be designed is required to transport the remote maintenance equipment (RME) near the desired Outer Cryostat surface, deploy inspection probes, and retrieve it once the task is completed. These preliminary requirements are decomposed into a detailed sequence diagram, where each sequence corresponds to a high-level transition in the SPN, as shown in Fig. 8 (Outer cryostat ribbons and under cryostat skirts – highlighted parts - are the stiffening structure of the Cryostat that need to be inspected.) The CAD model only illustrates the early-stage process. The Automated Guided Vehicles (AGV) in the image reflect a high-level concept, not the final system design.

Table 1 Input parameters for the design problem

Design Input	Specifications	Preliminary Considerations			
		AGV Concept		Rail-based Concept	
		Min	Max	Min	Max
Design parameters	Transportation speed v_{trans} (m/s)	0.833	1.8	0.3	0.833
	Deployment speed v_{dep} (m/s)	6.5	7	6.5	7
	Inspection speed v_{scan} (m/s)	0.1	0.3	0.1	0.3
	Deployment height of the outer Cryostat ribbons $h_{CSribbon}$ (m)	14.385 – 28.77			
	Deployment height in the under Cryostat Skirt $h_{CSskirt}$ (m)	4.763 – 6.99			
Uncertainty parameters	System failure probability	p			
	Time to repair (h)	1 – 18.3			

The design problem is formulated as a mathematical problem to find suitable design parameters to fulfil requirements and optimise the system's performance under QR (Table 1). Design and uncertainty parameters were inputs for the problem and are defined within a range of minimum and maximum values. Design parameter ranges are derived from the capabilities of similar existing products in the market, such as AGV or the rail-transfer cart. These preliminary considerations are obtained through feasibility studies and the documentation of relevant market systems based on the design-by-cataloguing approach (Truong & Brace, 2025). On the other hand, design uncertainty parameters represent aspects of worst-case scenarios, such as the system's failure (probability) in the working shift and the delayed time for the system's self-recovery activities. The system failure probability is generalised as a symbolic value p.

4.2. From Sequence Diagram to Stochastic Petri Net

For each high-level stochastic transition in Fig. 8, a minimum and maximum transition time is determined based on the input parameters from Table 1. Given a new range value for each transition, an equivalent distribution is estimated using the three-sigma rule, which models a standard process through a Normal distribution. The transition t13 is introduced as a worst-case scenario (system failure); typical failures are assumed to occur predominantly within 1 hour, while rare and severe downtime events may extend up to 18.3 hours. Transitions t1–t12 are modelled with normal distributions, reflecting the consistency of repetitive, automated tasks. In contrast, the recovery process in transition t13 is captured by a lognormal distribution, accounting for its variability and potential for extended delays. Table 2 shows the stochastic transition and equivalent distributions under hour units. For example, for transition t_1 in Table 2, the min 0.00238 and max transition time is 0.00514 hours, computed by dividing the transport distance of 15.414 meters

(measured from the plant's CAD layout) by the maximum and minimum transportation speed (v_{trans}) of 1.833 and 0.833 m/s, respectively, and converting the result into hours. This range is then approximated by a normal distribution with the mean set to the midpoint and the standard deviation defined as one-sixth of the range, in accordance with the three-sigma rule.

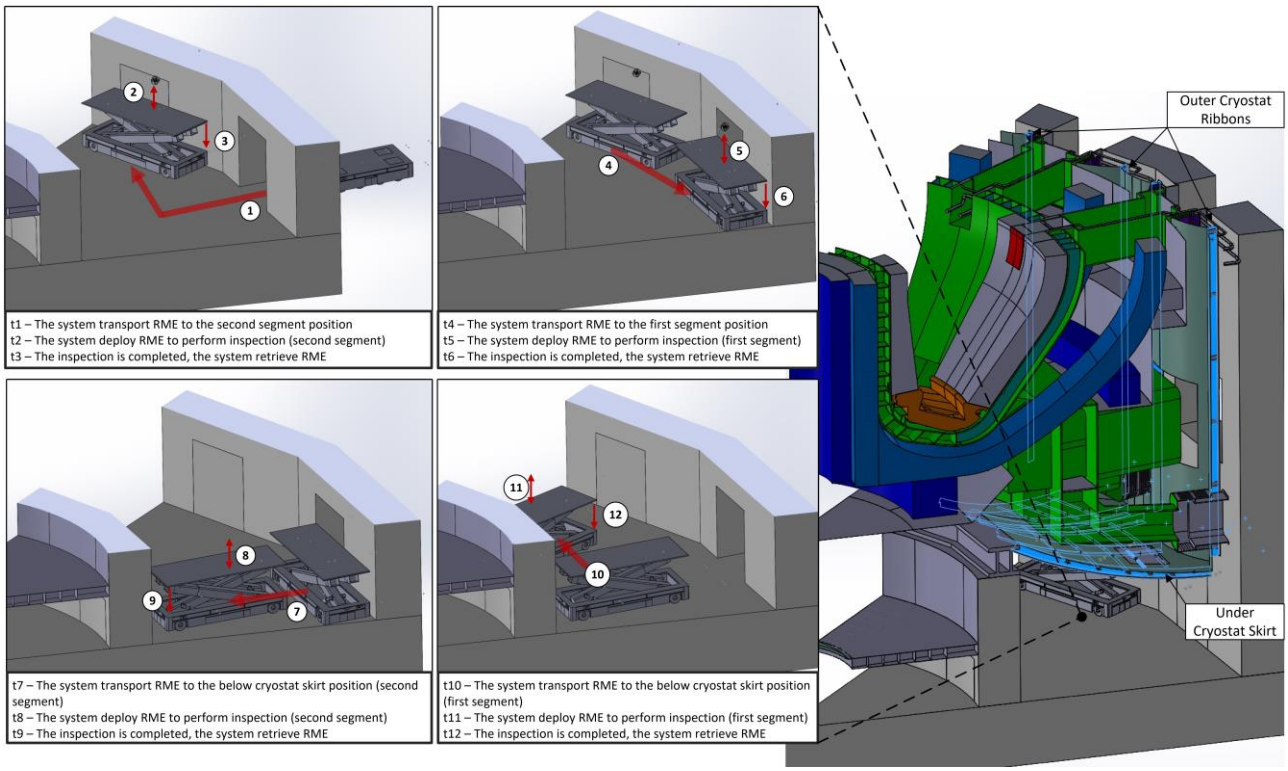


Fig. 8 Preliminary sequence diagram and system's tasks corresponding to high-level stochastic transitions from t1 to t12 (half-section view of the cylindrical In-BS region, showing 2 out of 16 segments)

Table 2 Stochastic transitions and equivalent distributions

Transition	Description	Min (hour)	Max (hour)	Equivalent Distribution (hour)
t1	Transportation Time, calculated by a fixed distance and a velocity range ($v_{trans\text{-}min}$ and $v_{trans\text{-}max}$)	0.00238	0.00514	$\sim\mathcal{N}(0.0036, 0.0046^2)$
t2	Deployment for inspection time. Depends on a fixed ultrasonic scanning speed and a scanning system range ($d_{height\text{-}min}$ and $d_{height\text{-}max}$)	0.03075	0.06150	$\sim\mathcal{N}(0.04615, 0.00512^2)$
t3	Retrieval time depends on a range of distance and range of deployment speed	0.00057	0.00123	$\sim\mathcal{N}(0.0009, 0.00018^2)$
t4	Transportation Time, calculated by a fixed distance and a velocity range (v_{min} and v_{max})	0.00137	0.00295	$\sim\mathcal{N}(0.00216, 0.00039^2)$
t5	Deployment for inspection time. Depends on a fixed ultrasonic scanning speed and a scanning system range ($d_{height\text{-}min}$ and $d_{height\text{-}max}$)	0.03075	0.06150	$\sim\mathcal{N}(0.04615, 0.00512^2)$
t6	Retrieval time depends on a range of distance and range of deployment speed	0.00057	0.00123	$\sim\mathcal{N}(0.0009, 0.00018^2)$
t7	Transportation Time, calculated by a fixed distance and a velocity range ($v_{trans\text{-}min}$ and $v_{trans\text{-}max}$)	0.00032	0.00070	$\sim\mathcal{N}(0.00051, 0.0001^2)$
t8	Deployment for inspection time. Depends on a fixed ultrasonic scanning speed (v_{scan}) and a scanning system range (d_{min} and d_{max})	0.13264	0.15274	$\sim\mathcal{N}(0.1427, 0.00337^2)$
t9	Retrieval time depends on a range of distance and range of deployment speed	0.00019	0.00030	$\sim\mathcal{N}(0.000245, 0.00019^2)$
t10	Transportation Time, calculated by a fixed distance and a velocity range (vmin and vmax)	0.00101	0.00217	$\sim\mathcal{N}(0.00159, 0.00029^2)$
t11	Deployment for inspection time. Depends on a fixed ultrasonic scanning speed and a scanning system range (d_{min} and d_{max})	0.13264	0.15274	$\sim\mathcal{N}(0.14270, 0.00337^2)$
t12	Retrieval time depends on the range of distance and deployment speed	0.00019	0.00030	$\sim\mathcal{N}(0.000245, 0.000019^2)$
t13	Failure distribution, with the mean of 1 hour of troubleshooting and a maximum of around 18.3 hours	0.0129	18.3	$\sim LN(-0.724, 1.21^2)$

A closed SPN network was constructed using the given distributions in Table 2, as shown in Fig. 9. This network is then modelled using the PNlib library in Modelica language. The "tSystem" transition in this network is a grouped transition, similar to a small subnet consisting of:

- A regular RM process block (transitions t1-t12) for the Sensitivity-based Analysis

- A failure block (failure branch with probability and delayed recovery time) for the performance-based Analysis

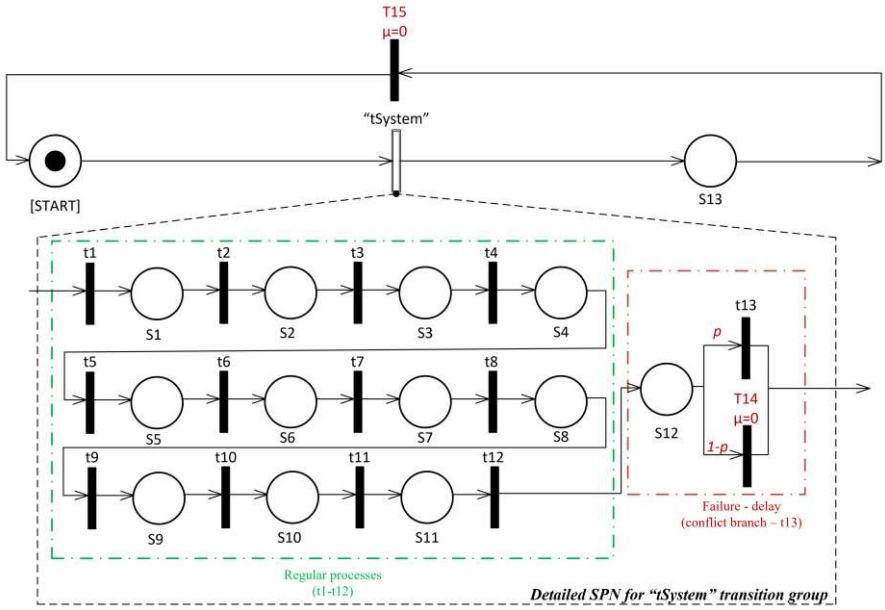


Fig. 9 Constructed closed SPN network, where "tSystem" is a transition group comprising regular processes and a conflict branch representing system failure with a probability of p.

It is noted that this network, based on PNlib, also includes immediate transitions as buffers to enable branching and lopping for the Monte Carlo simulations (transitions T14 and T15). The use of PNlib in Modelica offers advantages and flexibility in the early conceptual design due to the open-source nature of the library. This strength allows for modifications such as incorporating normal and lognormal distribution functions into the original library.

4.3. Sensitivity-based Analysis

The design input parameters are described as an input vector $X = (v_{trans}, v_{dep}, v_{scan}, h_{CSribbon}, h_{CSskirt})$. The design problem is formulated using equation (5) as a black box function of the input:

$$Y = f(X) = f(v_{trans}, v_{dep}, v_{scan}, h_{CSribbon}, h_{CSskirt}) \tag{8}$$

The input vector includes design and uncertainties derived from the requirements in Table 1. In this sensitivity-based analysis, the focus is on examining the crux of the process (regular RM process group in Fig. 9; therefore, the failure probability is set to zero ($p = 0$). The SA was conducted using the Sobol technique (a variance-based SA method based on equation (6)), implemented through the SALib library in Python, with a sample size $N = 16384$ for the sampling function. Sobol sampling was used to generate input parameter values, and the first-order Sobol index (S_i) and total-effect Sobol index (S_T) were calculated based on the normalised data to quantify the impact of each parameter on the overall process time.

Table 3 Sensitivity indices (Sobol) between design input parameters and total overall process time (highest to lowest)

Input parameters	S_i (Sobol')	S_T (Sobol')
v_{scan}	0.9752	0.9774
$h_{CSribbon}$	0.0223	0.0244
v_{trans}	$3.42 \cdot 10^{-4}$	$3.42 \cdot 10^{-4}$
v_{dep}	$5.06 \cdot 10^{-4}$	$5.11 \cdot 10^{-4}$
$h_{CSskirt}$	$9.37 \cdot 10^{-8}$	$8.97 \cdot 10^{-8}$

The sensitivity indices results are demonstrated in Table 3, with the input parameters normalised before the analysis. The results show that the overall process time of the design system is affected mostly by the inspection speed v_{scan}, where both first-order (S_i) and total (S_T) indice are high. This indicates that in the process, the majority of the effect comes from the direct contribution of the v_{scan} to the output of the design problem and the interaction effect between input variables are neglectable.

Since the overall process time is defined as a simple linear sum of the twelve sub-transition times, the contribution of each transition to the output variance is identical and theoretically fixed at $S_T = 1/12 \approx 0.083$. In such a purely additive case, running a full Sobol analysis at the process or transition level is unnecessary. However, if the process were to include branching paths (conflicts), a comprehensive sensitivity analysis would again be required to evaluate the relative

importance of each sub-transition. As a result, the concept design strategy of the system should not focus on improving the transportation speed, such as considering the use of either AGV or rail-based systems (preliminary concepts in Table 1). Instead, efforts should be prioritised to improve the system's inspection speed v_{scan}.

4.4. Performance-based Analysis

This performance-based analysis focuses on the impact of worst-case scenarios on the system through uncertainty parameters (system failure probability and time to repair in Table 1). This is done by enabling the "failure-delay" block in Fig. 9 b) by granting a value of conflict branching probability $p \neq 0$. This SPN is equivalently transformed into the irreducible Markov chain, as illustrated in Fig. 10, with the stochastic matrix of;

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & p & 0 & 1-p \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{9}$$

Solving steady-state probabilities as a result of $\pi_{S1.1} = \pi_{S1.2} = \pi_{S1.5} = \frac{1}{2p+3}, \pi_{S1.3} = \pi_{S1.4} = \frac{p}{2p+3}$.

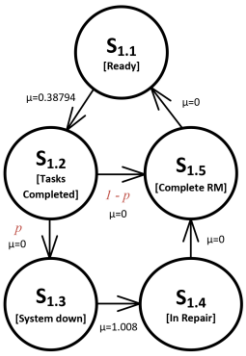
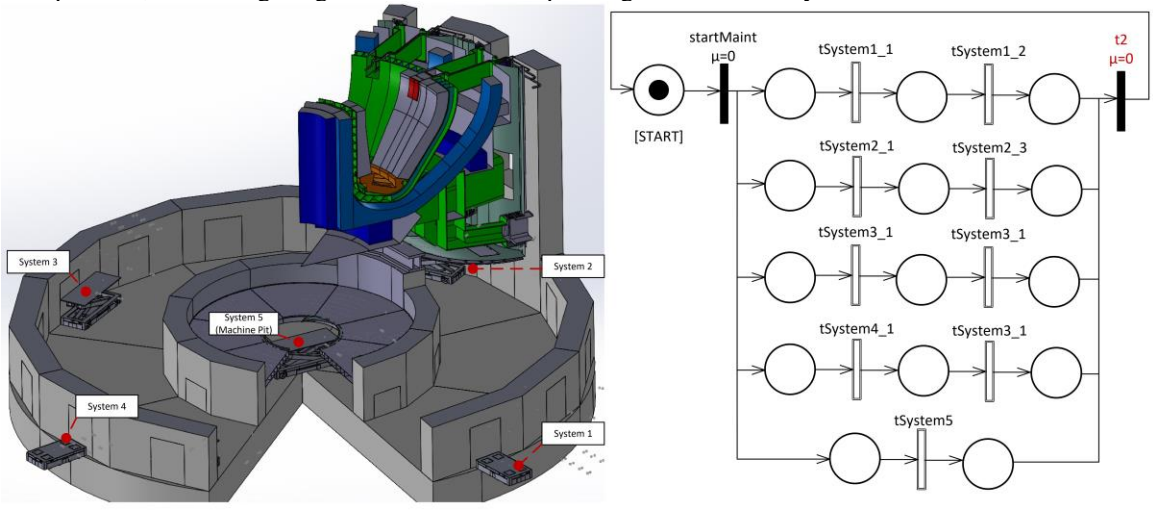


Fig. 10 Markov chain for one single system (unbroadened problem)

With these limiting probabilities, the process's overall mean time is determined by using equation (2):

$$\mu_{single} = \frac{1}{d}\sum_{\substack{i=1, \\ i\neq e}}^{m} \pi_i \mu_i = (2p+3)\left(\frac{0.38794}{2p+3} + \frac{p}{2p+3}1.008\right) = 1.008p + 0.38794\ (hour) \tag{10}$$

It can be seen that the overall execution time μ_{single} follows a simple linear function, and the steady-state probabilities are basic functions of p. Thus, using a single system to perform RM for the In-BS is impractical, given that this large area has an estimated diameter of 44 m and a height of 42 m. Therefore, to overcome this problem, the design problem is broadened by integrating multiple parallel systems performing RM simultaneously. This expansion introduces additional complexity but allows for the effective use of SM models to solve problems. Fig. 11 - left illustrates the sequence for this broadened problem, and the right figure shows the corresponding SPN network by Modelica.



a) Extended problem (5 concurrent systems) *b) SPN model of 5 concurrent systems*

Fig. 11 Broadened problem using 5 concurrent systems (left) and the equivalent SPN model (right)

Consider the total system is a set of interactions between five identical individual systems; in such a case, each system has an identical failure probability p, then the probability of having exactly k $(0 \leq k \leq 5)$ failed systems out of the five will follow the common-known probability mass function of the binomial distribution $p_{k\ systems} = \binom{5}{k} p^k (1-p)^{5-k}$. The equivalent SM network is obtained by applying SM modelling with performance-based analysis, as shown in Fig. 12. This irreducible Markov chain represents the behaviour of the overall system, with five potential failure states, indicating how many subsystems may fail at the same time, from one up to five ($S_{2.3} - S_{2.7}$).

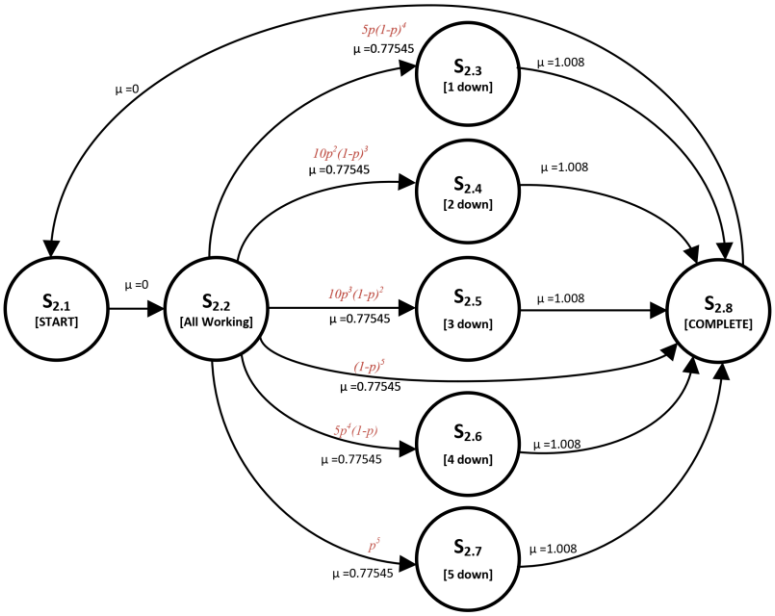


Fig. 12 SM model for the broadened problem.

The stochastic matrix for this broadened network now takes the following form:

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5p(1-p)^4 & 10p^2(1-p)^3 & 10p^3(1-p)^2 & 5p^4(1-p) & p^5 & (1-p)^5 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (11)$$

From this stochastic matrix, the steady-state probability values can be derived as $\pi_{S2.1} = \pi_{S2.2} = \pi_{S2.8} = \frac{1}{4-(1-p)^5}, \pi_{S2.3} = \frac{5p(1-p)^4}{4-(1-p)^5}, \pi_{S2.4} = \frac{10p^2(1-p)^3}{4-(1-p)^5}, \pi_{S2.5} = \frac{10p^3(1-p)^2}{4-(1-p)^5}, \pi_{S2.6} = \frac{5p^4(1-p)}{4-(1-p)^5}, \pi_{S2.7} = \frac{p^5}{4-(1-p)^5}$. By utilising equation (2), the functional relationship between the process's overall execution in terms of the input failure probability of the system can be obtained as:

$$\mu_{concurrent} = \frac{1}{d}\sum_{\substack{i=1, \\ i\neq e}}^{m} \pi_i \mu_i = \frac{1}{\pi_{S2.1}}[\pi_{S2.2} 0.77545 + 1.008 \sum_{i=S2.3}^{i=S2.7} \pi_i] = 0.77545 + 1.008(1-(1-p)^5)(hour) \quad (12)$$

Results from equations (11) and (13) show that the extended problem has become a fifth-degree nonlinear function, which is more complex than the single system case in Fig. 10. This demonstrates the dependence of the limiting probabilities and the overall process time on the p value. Fig. 13 (left) shows the plot of steady-state probabilities when p increases. In this figure, it is evident that when the system's failure probability $p < 0.2$, the probability of two systems failing at the same time remains low ($\pi_{S2.4}$), while the likelihood of three to five systems operating remains high. As p increases, the probability of stable systems' operation ($\pi_{S2.2}$) declines sharply and gradually converges to 0.25. Conversely, the probabilities of multiple system failures increase significantly. Fig. 13 shows the plot of equation (12) corresponding to p. Similarly to the left subplot of Fig. 13, it can be seen that failure probability is directly proportional to the system's overall process time due to the delays for system self-recovery.

At this stage, the impact of the uncertainty parameter p on the overall system behaviours becomes clearer. Therefore, it is essential to select a suitable limit of p as a design reliability specification. Since p represents the probability of a

single system's failure during a fixed continuous operation cycle T, it has a relationship with the Mean Time Between Failure (MTBF) through the following survival function (exponential distribution):

$$1 - p = e^{-\frac{T}{MTBF}} \tag{13}$$

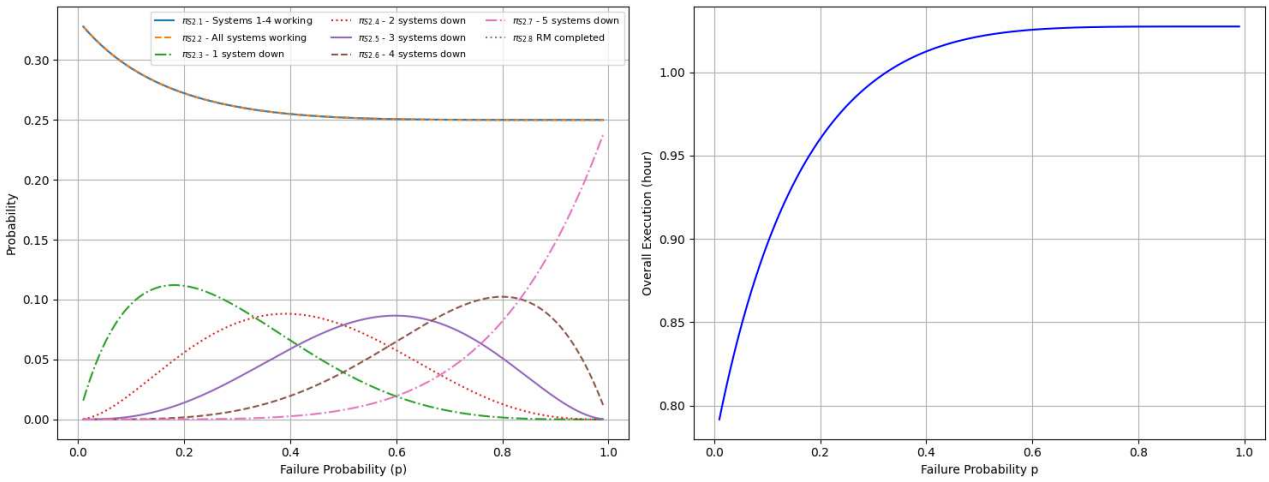


Fig. 13 Steady-state probabilities vs failure probability p (left) and the expectations of overall execution time vs p (right).

In the context of the design problem, preventive maintenance is assumed to be scheduled every 30 months, with each maintenance session lasting up to 3 months. By selecting $T = 3$, the dependency between MTBF (months) and the failure probability p can be illustrated in Fig. 14.

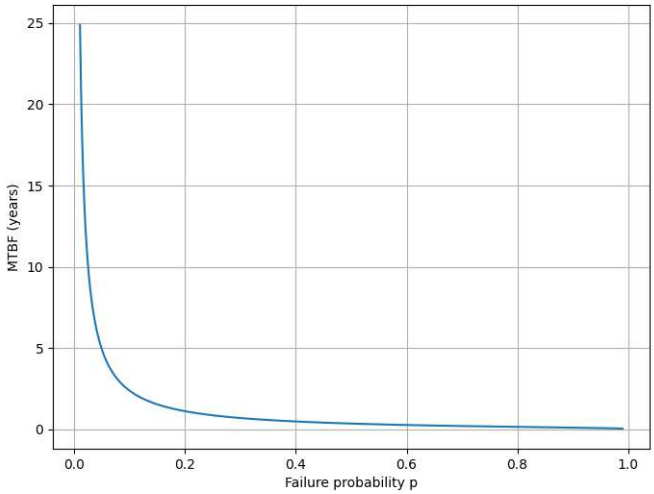


Fig. 14 Relationship between failure probability p and required MTBF (months) for design.

4.5. Risk Assessment through Uncertainty Analysis

Fig. 13 and Fig. 14 provide an overview and constraints for selecting an appropriate p value for the design reliability. Specifically, p should be selected below the peak value $p < 0.181$ of the $\pi_{S2.3} = 0.112$ in Fig. 13-left to prevent the probability of severe downtime of simultaneous subsystems. However, if p is too low, the required MTBF will be significantly high, raising concerns about design feasibility and associated cost (Fig. 14). In this case study, four pairs of values of p and v_{scan} were selected as four corresponding test plans for risk assessment, as listed in Table 4.

Table 4 Test Plan (design specifications) for Risk Assessment.

Configurations	**156_0.05**	**156_0.03**	**130_0.05**	**130_0.03**
$v_{scan}(mm/s)$	156	156	130	130
p	0.05	0.03	0.05	0.03
Equivalent design MTBF (years)	4.87	8.2	4.87	8.2

The value $v_{scan} = 130\ mm/s$ is used as a baseline value of the input range value presented in Table 1. The preferred high-level solutions aim to optimise the design by improving v_{scan} by 20%, corresponding to $156\ mm/s$, while

evaluating cases with $p = 0.03$ and $p = 0.05$ to assess their impact. The risk assessment was conducted using the Monte Carlo simulation of the SPN model in Fig. 11 based on these test plan values. The histogram and density plot results for each tech plan are demonstrated in Fig. 15 with observations $N = 199250$. It is noted that due to the long-tail nature of the distribution, x axes in the plots are restricted to median + 3σ to ensure the clear visualisation of the distribution shape.

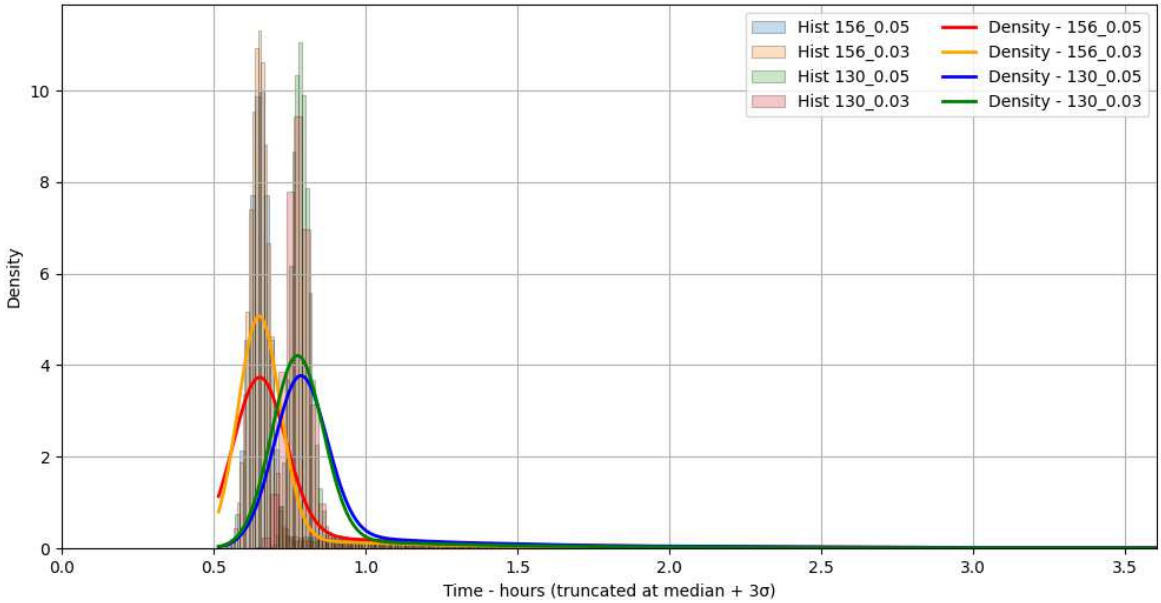


Fig. 15 Histogram with density plots of the system performance for the risk assessment between configurations

The histogram and density plot results indicate a difference in the distribution shape between $v_{scan} = 130\ mm/s$ and $v_{scan} = 156\ mm/s$, as v_{scan} increases, the distribution shifts further left. Thus, a higher p results in a lower peak and a thicker tail, showing a higher occurrence of worst-case events, increasing overall system risk.

The QR can be evaluated quantitatively using the Cumulative Density Function (CDF) and statistical information for each test configuration, as shown in Fig. 16 (left) and Table 5. The CDF curves of 4 configurations remain flat, showing no probability accumulated in the lower range, then rise sharply over a short interval, indicating most process time is concentrated within a specific range before gradually flattening and extending toward the upper limit, characteristics of a right-skewed distribution of the worst-case scenario (transition t13 with lognormal distribution).

Table 5 Statistics summary between 4 sets of design specifications and its trade-offs in equivalent design MTBF (observations $N = 199250$).

Statistics	**156_0.05**	**156_0.03**	**130_0.05**	**130_0.03**
Median	0.66	0.66	0.79	0.78
Mean	0.87	0.78	1	0.91
Stdev	0.94	0.71	0.94	0.87
Minimum	0.52	0.52	0.67	0.61
Maximum	48.19	30.52	33.29	79.21
Q1 (25% quantile)	0.63	0.63	0.77	0.75
Q3 (75% quantile)	0.7	0.68	0.83	0.81
Equivalent design MTBF (years)	4.87	8.2	4.87	8.2

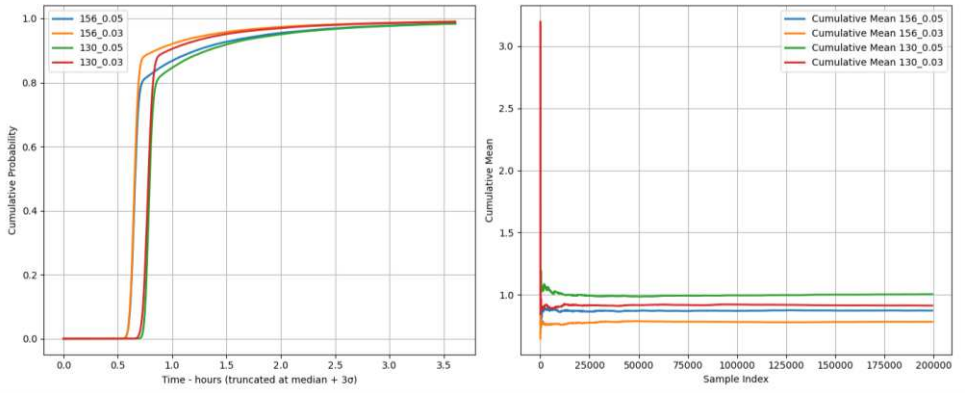


Fig. 16 Cumulative Density Function plot (left) and Cumulative Mean Convergence of Monte Carlo simulation (right)

The statistics summary in Table 5 reveals that the median value decreases by approximately 16.46% under the optimal configuration 156_0.03, compared to the baseline 130_0.05. This improvement was achieved with a 20% increase in scan speed and a 2% reduction in failure probability. Thus, as p increases, the standard deviation grows, and there is a shift in Q1 and Q3. The reliability of the Monte Carlo simulation is demonstrated in Fig. 16 (right), which shows that as the number of simulations increases, the system behaviour gradually stabilises and converges toward the expected value, aligning with the law of large numbers.

Table 6 shows the results of the pairwise Cliff's Delta for four test configurations. It is again clear that a 20% increase in v_{scan} leads to an improvements in system performance, with δ progressing from medium to large effects, from 0.65581 to 0.6789998. Conversely, when v_{scan} remains unchanged and only p is improve, the performance improvement is negligible, suggesting that the v_{scan} contribute more significant role in optimising the overall system performance.

Table 6 Pairwise Cliff's delta results for the test configurations

	156_0.05	**156_0.03**	**130_0.05**	**130_0.03**
156_0.05	0	0.079134	-0.678998	-0.655810
156_0.03	-0.079134	0	-0.802826	-0.785628
130_0.05	0.678998	0.802826	0	0.194123
130_0.03	0.655810	0.785628	-0.194123	0

The impact of selecting different distribution types for each task in the sub-transitions (as shown in Table 2) on the final outcome of a single system is illustrated in Fig. 17 (configuration 156_0.05 is used as an example). The orange PDF shows the completion time using non-Markovian assumptions, which are a combination of normal and lognormal distributions defined in Table 2. In contrast, the red PDF represents the results assuming all transitions follow an equivalent exponential distribution, which is the traditional Markovian SPNs. Under the exponential assumption, there is a high number of completion times near zero, even though the actual theoretical minimum should be the sum of the minimum durations of all sub-transitions in Table 2. This unrealistic result stems from the exponential distribution's support starting at zero, which causes an inflated probability of near-instant completions.

On the other hand, the non-exponential model yields a more realistic output, with the shortest completion time at 0.52 hours and most values clustered around the median of 0.66 hours. This pattern reflects repeated automation tasks (t1–t12) with variances modelled using normal distributions combined with lognormal distributions that represent rare but severe delays. These findings, combined with the earlier discussions in Section 2.2 on the limitations of exponential assumptions, show the inadequacy of exponential distributions in accurately representing real-world operational behaviours. Choosing suitable distribution assumptions at the early of RA enhances model realism and facilitates the accuracy of the predictive model.

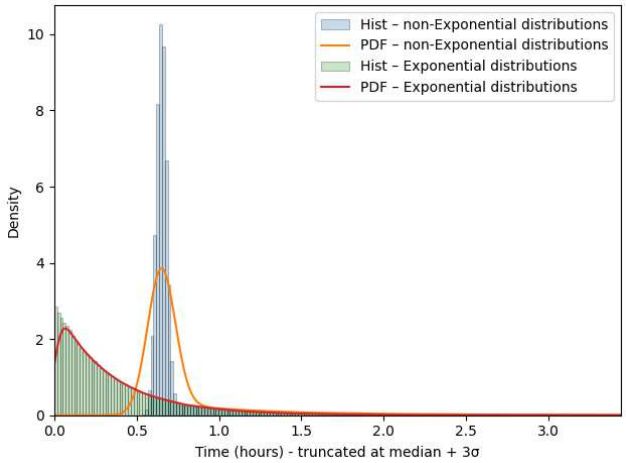


Fig. 17 Comparison results between exponential assumptions (Markovian SPN) and non-exponential assumptions (non-Markovian SPN) on the 156_0.05 specification

In summary, the statistical results from the risk assessment indicate that optimising the system's inspection speed (v_{scan}) as part of the conceptual design strategy, improves system QR significantly as a crux of the design problem. Furthermore, there is a strong correlation between failure probability p and the MTBF, which can be considered a key design specification for subsequent design stages. The outcomes of the sensitivity-based analysis indicated that the crux of the design problem is the inspection speed (v_{scan}) with the preferred design specifications under test configurations outlined in Table 4. The optimal combination of design specifications (preferred design specifications) can be determined using insights from Fig. 16, Table 5 and Table 6 while also considering the trade-off between design costs, feasibility and acceptable risk levels.

5. Conclusions

This article introduces a quantitative risk analysis in the early conceptual design stages to identify and mitigate the system's quality risks. The framework focuses on the initial problem-solving stage of the conceptual design by broadening and abstracting to find the crux of the problem. QR is defined as performance risks that might impact the system's ability to complete the overall task on time. The key idea of this method is to view design problems from a mathematical perspective, modelling system behaviour as a sequence of processes required to fulfil predefined tasks or requirements. Aside from this, the design problem is formulated as a black-box function, where the output, Y, is the system performance, and the inputs consist of design specification and uncertainty (characterising worst-case scenarios) parameters. The process modelling relies primarily on non-Markovian SPN representation, with the output conflicts used to capture worst-case scenarios.

The RA and assessment of the approach are built upon the integration of four key knowledge foundations:

- **SPN:** serves as the framework for modelling sequence diagrams under discrete events and facilitates the visualisation of these processes in the early design stage through directed graphs. The SPN allows for a flexible conversion to other mathematical models, such as irreducible Markov chains.
- **Sensitivity-based analysis:** Identify key design specifications and subprocesses (transitions) in the SPN network that significantly influence system performance. These critical factors represent the "design crux" regarding performance impact.
- **Performance-based analysis** expands the problem scope by examining uncertainty parameters and their interrelationships to system performance. This analysis is based on the irreducible Markov chain derived from the original SPN network.
- **Uncertainty analysis:** integrates findings from SA and performance analyses to determine acceptable values for design specifications and uncertainty parameters that minimise QR through Monte Carlo simulations. Decision-making at this stage is guided by statistical analysis of the output variable Y for each simulated set of input variables.

A case study on the design of an RM deployment system for the In-BS area of the EU-DEMO fusion power plant was presented to demonstrate the practical application of the proposed method. The results highlighted the strong compatibility of using SPN in the early stages of the conceptual design due to its strength in modelling discrete events. As directed graphs, it provided a highly visual modelling interface and facilitated the representation of discrete events through modular connections of transitions and places. This structure supports intuitive system representation and is compatible with open-source tools like OpenModelica using PNlib. Moreover, transforming SPNs into closed networks enables integration with various statistical analysis methods. Compared to traditional Petri Net modelling approaches, SPNs offered greater flexibility, as transitions can be defined as random variables to overcome the limitation of fixed or predefined transition times while still being capable of describing worst-case events through output conflict representation.

The results demonstrated that various statistical methods can be applied to the proposed design method, particularly in sensitivity-based analysis or risk assessment of Monte Carlo simulation outcomes. The choice of method depends on the desired accuracy. When the analysis is not deep and practical enough, it can be broadened, where semi-Markov models and Monte Carlo simulations help to quantify these complex issues. Risk analysis results can be quantified through UA. At the same time, high-level solutions based on the identified crux design parameters can be determined by comparing distributions using techniques such as pairwise Cliff's Delta or statistical summaries. This method provides different perspectives on the probabilistic risk tolerance of the system, allowing its translation into design requirements as a feedback loop (e.g., MTBF in this case study). It is important to recognise that the most favourable uncertainty and design parameter values are not always the best choices, as they must be considered against the design feasibility and associated cost trade-offs. Therefore, different high-level solutions can be selected depending on risk appetite to balance system reliability, performance and cost.

Overall, the proposed method provided a quantitative approach to identifying how QR affects the system design in the early conceptual design stages. One of its key strengths is its ability to identify crux processes design parameters, assess risks, and examine system behaviours based on the design context, even when the system's physical structure has not yet been fully defined. While the method shows strong potential, certain practical considerations remain. The computational cost is relatively high, as it heavily relies on Monte Carlo simulations, particularly for systems with a high number of concurrent transitions (e.g., above 15 transitions). This aspect highlights potential directions for future research, such as improving computational efficiency through state-space reduction strategies (e.g., marking aggregation or fluid approximations) and refining parameter estimation methods.

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